Pre-Analysis Plan

#ManyDaughters

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1. Hypotheses to test

H1. *Having daughters leads to stronger support for gender equality (as measured by attitudes)*.

That is, denoting X as a measure of support for gender equality,

(X for respondents with daughters) > (X for respondents without daughters).

H2. *Having daughters leads to more left-leaning political preferences*.

That is, denoting Y as a measure of left-leaning political preferences,

(Y for respondents with daughters) > (Y for respondents without daughters).

H3. *Having daughters leads to more gender equality in the household (as measured by behaviors or outcomes)*.

That is, denoting Z as a measure of gender equality in the household,

(Z for respondents with daughters) > (Z for respondents without daughters).

H4. *Having daughters leads to a larger increase in the support of gender equality for men than for women (with gender equality measured by attitudes)*.

That is, denoting X as a measure of support for gender equality,

(X for male respondents with daughters − X for male respondents without daughters) > (X for female respondents with daughters − X for female respondents without daughters)

2. Variables

This section outlines the definitions and construction of the variables used in this analysis. We begin by describing how we construct the control variables and the variables required for identification. We then turn to the construction of the outcome variables. While the control and identification variables remain fix throughout the analysis, the outcome variables vary across hypotheses.

Identification variables

We construct two variables needed for identification.

$HasDaughter_{\mathrm{i}}$

We construct a binary variable $HasDaughter_{\mathrm{i}}$ that equals 1 if individual *i* has at least one daughter and 0 otherwise. Specifically, the variable is coded as 1 if any of the variables kidsex01 to kidsex19 indicate a female child.

$FirstChildIsDaughter_{\mathrm{i}}$

We construct a binary variable $FirstChildIsDaughter_{\mathrm{i}}$ that equals 1 if the first child (kidsex01) of individual *i* is female and 0 otherwise.

Control variables

We use the following control variables: male parent, respondent age, respondent parental age, respondent's years of education, respondent's income (quartile), year fixed effects.

$Male_parent_{\mathrm{i}}$

We construct a binary variable $Male_parent_{\mathrm{i}}$ that equals 1 if individual *i* variable sex is coded as 1 and 0 if the variable is coded as 2.

$Number_kids_{\mathrm{i}}$

We construct the variable $Number_kids_{\mathrm{i}}$ that indicates the number of kids.

Age_{i}

We construct the variable Age_{i} that is defined as 2025 minus the variable gebjahr. The variable gebjahr captures the birthyear of individual i.

$Parental_age_{\mathrm{i}}$

We construct the variable $Parental_age_{\mathrm{i}}$ that is defined as 2025 minus the variable kidgeb01. The variable kidgeb01 captures the birthyear of individual i's first child.

$Year_edu_{\mathrm{i}}$

We construct the variable $Year_edu_{\mathrm{i}}$ that corresponds to the variable pgbilzeit and captures individual i's years in education.

$Income_groups_{\mathrm{i}}$

We construct the variable $Income_groups_{\mathrm{i}}$, which indicates the income quartile of individual *i* in year *t*. Income is defined as the sum of gross labor income (labor_inc) and secondary income (secondary_inc).

$Year_{\mathrm{i}}$

We construct the variable $Year_{\mathrm{i}}$ that corresponds to the variable syear that captures year in which the survey was conducted.

$Region_{\mathrm{i}}$

We construct the variable $Region_{\mathrm{i}}$ that corresponds to the variable sampreg that captures if a respondent is in the sample for East or West Germany. We use this variable in absence of any other residence indicators.

Outcome variables

$GenderAttitude_i$

As a dependent variable for Hypothesis 1 and 4, we construct a composite variable that capture gender attitudes. We include the following variables: plj0620, plj0621, plj0622, plj0623, plj0624, and plj0625. At the variables have different scales (plj0620: 1-10, plj0621- plj0625: 1-7) and different directions, we first transform them the following way.

First, we recode the variables plj0623, plj0624, and plj0625 such that higher number indicate attitudes in line with gender equality. Specifically, we switch the label the following way 1 -> 7, 2 -> 6, 3->4, ..., 7->1.

Second, we bring all variable to a comparable scale by normalize all variables to zero mean and unit variance.

Third, we compute the variable $GenderAttitude_i$ as individual i's average over all standardized variables.

$PoliticalLeaning_i$

As a dependent variable for Hypothesis 2, we construct the variable $PoliticalLeaning_i$. The variable corresponds to the variable plh0004 that measures self-reported left-right leaning of individuals. It is measures on a scale from 0 to 10, where 0 indicates far left and 10 indicates far right.

$Homework_i$

As the dependent variable for Hypothesis 3, we use the variable $Housework_difference_i$ that corresponds to the absolute difference of weekly hours that couples within a household spend on household chores and childcare.

We first construct the variable $Homework_i$, which captures the number of hours per week that individual *i* spends on household chores and childcare. The variable is constructed as the sum of reported hours spent on these activities during weekdays and weekends: $Homework_i$ = *pli0019_h_clean* + *pli0022_h_clean* + *pli0044_h_clean* × 5 + *pli0012_h_clean* + *pli0016_h_clean* + *pli0043_h_clean* × 5.

To ensure plausibility, we winsorize the variable at the maximum possible number of hours per week (168). That is, individuals who report spending more than 168 hours per week on household chores and childcare are assigned a value of 168.

We then compute the $Housework_difference_i$ that corresponds to the absolute difference of the variable $Homework_i$ of couples within a household in year t. Zero implies that both individuals spend the same amount of weekly hours on household chores and child care. Increasing values indicate increasing inequality.

3. Inclusion/exclusion criteria

Parents

In our analysis, we compare the effects of having daughters as compared to having sons. Thus, effectively we are limiting our sample to parents.

Parental age larger 10

We measure gender attitudes starting from 10 years after the birth of the first child to capture a deeply internalized, experience-driven change, rather than a short-term emotional reaction. This longer time horizon is more likely to reflect the accumulated influence of real-life experiences with a daughter, including exposure to gender-based discrimination in school, during adolescence, or in early labor market interactions. By focusing on long-run attitudes, we aim to identify more durable and meaningful shifts in beliefs about gender equality

Missing observation/ data availability

Apart from these we will use all observations available in the data. However, data availability (not all questions are available every year) as well as item and unit non-response influence the sample composition which will be different our outcome variables.

4. Exact analysis

We will establish a causal relationship by exploiting the gender of the first-born child. Specifically, we use having a daughter as the first-born as an instrument for having at least one daughter. The identification strategy (first stage) is the same across all hypotheses. Specifically, we will estimate the following regression:

First stage:

$$HasDaughter_{\mathrm{i}} = \pi_0 + \pi_1 \cdot FirstChildIsDaughter_{\mathrm{i}} + X_{\mathrm{i}}'\pi + u_{\mathrm{i}}$$

In the second stage, we use the predicted probability from the first stage to estimate the effect of having at least one daughter on various outcome variables. The second-stage specification

differs between hypotheses H1–H3 and H4. For H1–H3, we estimate the direct effect of having daughters on the outcomes. In contrast, for H4, we examine whether this effect differs by the gender of the respondent. To do so, we interact the predicted probability of having at least one daughter with a gender indicator. Specifically, we estimate the following models.

Second stage (H1-H3):

$$Outcome_i = \alpha + \beta \cdot \widehat{HasDaughter}_i + X_i'\gamma + \varepsilon_i$$

Second stage (H4):

$$Outcome_i = \alpha + \beta \cdot \widehat{HasDaughter}_i \cdot Father + \theta \cdot \widehat{HasDaughter}_i + \vartheta \cdot Father + X_i'\gamma + \varepsilon_i$$

Where:

- $HasDaughter_i$ is 1 if Individual i has at least one daughter and 0 otherwise
- $FirstChildIsDaughter_i$ is 1 if Individual i has at least one daughter and 0 otherwise
- $Outcome_i$ is the outcome variable of interests. It varies with each hypothesis:
    - H1+H4: $GenderAttitude_i$ represents Person i's attitudes toward gender equality
    - H2: $PoliticalLeaning_i$ represents Person i's political leaning (0 = left, 10 = right)
    - H3: $Housework_difference_i$ represents the absolute difference of weekly hours spent on household chores and childcare between couples in household i.
- $\widehat{HasDaughter}_i$ is estimated probability of having at least 1 daughter
- X_i : Covariates (gender, age, parental age (age at first child), education (years of education), income (quartiles), year fixed effects, region, number of kids)
- β: The causal effect of interest

5. Interpretation of results

To test our hypotheses, we will rely on t-tests of the estimated coefficients in our regression models. Specifically, we will examine the p-values associated with the relevant coefficients to assess whether the results are consistent with our hypotheses. A result is considered statistically significant—and thus in line with the hypothesis—if the p-value is below our chosen significance threshold of 0.05. This implies that we reject the null hypothesis of no effect at the 5% significance level.